

Gihan Ariyasena

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Education

University of Staffordshire

Bachelor of Science in Computer Science

(2025-2028)

Experience

Software Developer Intern | Founder Diary | founderdiary.in | Remote

(Jul - Sep 2026)

- Contributed primarily to backend development on projects assigned by the team, building REST API endpoints and server-side logic.
- Worked on AI implementation across assigned projects, integrating AI features and capabilities into the applications.
- Built and connected frontend components to backend services for new features, handling end to end API to UI integration.

Projects

UMAP Algorithm From Scratch | Python, Jupyter Notebook

(Mar 2026)

- Implemented UMAP fully from scratch in pure NumPy, covering pairwise distance computation, kNN graph construction (k=15), binary search sigma calibration, fuzzy membership symmetrization, and cross entropy loss optimization over 200 epochs with no third party UMAP library used.
- Applied to the Wholesale Customers Dataset (440 samples, 6 features), compressing 6D data to a 2D embedding via custom PCA initialization that captured 72.46% of cumulative variance across the first two principal components.
- Repository: github.com/Gihan526/umap-algorithm-from-scratch

Shell Chat | Bun, TypeScript, OpenTUI, TanStack Router, Socket.IO

(Feb 2026)

- Built a real time terminal chat app with a home screen and private chat rooms, using Socket.IO for bidirectional messaging and OpenTUI for a fully keyboard driven TUI with no browser required.
- Structured as a Bun TypeScript monorepo with isolated web/ and server/ workspaces, with a single bun run dev:all command spinning up both services concurrently.
- Used TanStack Router's file based routing to auto generate routes from the file structure, so adding a new screen requires no manual route registration.
- Repository: github.com/Gihan526/shell-chat

Achievements

Most Innovative Concept | Hemas Althon

(Apr 2026)

Team QuadStack | Project: InsightQL

AI powered natural language interface for enterprise data. Users query databases in plain English and instantly receive interactive dashboards, eliminating manual chart building.

1st Place | APIIT Kandy Hackathon

(Jul 2025)

Team Optiq | Code Verse 2025

Built a RAG study platform where students upload materials, ask questions, and get auto-generated quizzes.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL (PostgreSQL, MySQL), Bash, Rust

Frameworks & Libraries: React.js, Next.js, Node.js, Express, Tailwind CSS, Socket.IO

AI/ML & Data: Pandas, Scikit-learn

Databases: PostgreSQL, MySQL, SQLite, Redis

Certifications

AWS Cloud Practitioner Essentials

(Jul 2026)